

Data Analysis Midterm Project

Project Overview

The objective of this project is to apply the complete data analysis workflow to a real-world dataset. You will select an appropriate dataset, prepare it for analysis, explore and analyze the data, and communicate your findings through an interactive Power BI dashboard.

This project is designed to assess your ability to apply the concepts and tools covered throughout the course while demonstrating analytical thinking and effective data storytelling.

AI Usage Policy

AI tools (such as ChatGPT, Microsoft Copilot, Gemini, or similar tools) may be used to assist your work. However, AI should support your learning—not replace it. You are responsible for understanding every step of your project and verifying the accuracy of all AI-generated content.

Project Workflow

Part 1 — Dataset Selection

Select a dataset that satisfies **all** of the following requirements.

General Requirements

The dataset must:

- Be a **structured tabular dataset** (organized into rows and columns).
- Be related to a real-world domain such as:
 - Sales
 - Finance
 - Human Resources (HR)
 - Marketing
 - Healthcare
 - Education
 - Customer Analytics
 - E-commerce
- Contain **at least 1,000 records (rows)**.
- Contain **at least 8 variables (columns)**.

Data Characteristics

The dataset should:

- Include both **numerical** and **categorical** variables.
- Contain meaningful relationships between variables.
- Allow grouping, filtering, aggregation, and comparison.
- Include one or more measurable business metrics (e.g., Sales, Revenue, Profit, Salary, Quantity Sold, Customer Satisfaction, Exam Score, etc.).

Data Quality

The dataset should contain some real-world imperfections, such as:

- Missing values.
- Duplicate records.
- Incorrect or inconsistent values.
- Incorrect data types.
- Outliers or unusual observations.

These issues provide an opportunity to demonstrate data cleaning and preprocessing skills.

Analysis Potential

The dataset should allow you to:

- Perform meaningful exploratory data analysis (EDA).
- Answer business or analytical questions.
- Discover patterns and trends.
- Generate useful business insights.
- Create informative dashboards and visualizations.

Dataset Justification

In your report, explain:

- Why you selected this dataset.
- What business problem or analytical objective it represents.
- Why it is suitable for this project.

Part 2 — Data Preparation

Using **Microsoft Excel**, clean and prepare your dataset for analysis.

Where applicable, your cleaning process should include:

- Handling missing values.
- Removing duplicate records.
- Correcting inconsistent values.
- Fixing incorrect data types.
- Standardizing text and formatting.
- Identifying data quality issues that cannot be corrected.

Document all major cleaning steps in your report.

Part 3 — Exploratory Data Analysis (EDA)

Analyze the dataset to understand its characteristics and discover meaningful insights.

Your analysis may include:

- Summary statistics.
- Distribution of variables.
- Category comparisons.
- Trend analysis.
- Grouping and aggregation.
- Relationships between variables.
- Outlier detection.
- Business observations and insights.

Your goal is to answer meaningful questions about the data rather than simply creating charts.

Part 4 — Dashboard Development

Using **Power BI**, create an interactive dashboard that effectively communicates the insights from your analysis.

Your dashboard should:

- Present key findings clearly.
- Include appropriate visualizations.
- Include interactive slicers and filters.
- Use meaningful titles and labels.
- Follow good dashboard design principles.
- Tell a clear story supported by the data.

Choose visualizations that best communicate your findings rather than trying to use every available chart type.

KPI Requirements

Your dashboard must include **at least one Key Performance Indicator (KPI)**.

For each KPI, you must:

- Clearly define what the KPI measures.
- Define a **target value** for comparison.
- Explain **how the target was determined**.
- Justify **why the target is appropriate** for the business scenario.
- Interpret whether the KPI met, exceeded, or fell below the target.

Note: *A KPI is a metric that measures progress toward an important business objective. For this project, every KPI **must** be compared against a clearly defined target or benchmark. The target provides context, allowing you to determine whether performance is meeting expectations, exceeding them, or falling short.*

Deliverables

Submit **all** of the following files.

1. Dataset Description (PDF)

Provide a brief description of the selected dataset.

The document should include:

- Dataset title.
- Dataset source.
- Business domain.
- Number of rows and columns.
- Description of each variable.
- Why you selected the dataset.
- Business problem or analytical objective.
- Why the dataset satisfies the project requirements.

2. Cleaned Dataset (.xlsx)

Submit the cleaned Excel workbook used throughout the project.

The workbook should include:

- The cleaned dataset.
- All cleaning reflected in the final data.
- Appropriate formatting.
- No unnecessary worksheets or temporary calculations.

3. Power BI Dashboard (.pbix)

Submit the Power BI project file containing your dashboard.

The dashboard should include:

- Meaningful visualizations.
- At least one KPI measured against a target.
- Slicers and interactive filters.
- Clear titles and labels.
- Consistent formatting and layout.
- A professional appearance.
- Insights that help answer the business questions.

4. Project Report (PDF)

Prepare a report (approximately **2–4 pages**) summarizing your work.

Your report should include the following sections.

1. Dataset Overview

- Dataset description.
- Business domain.
- Objectives of the analysis.
- Justification for selecting the dataset.

2. Data Cleaning

Describe:

- Data quality issues identified.
- Cleaning techniques performed.
- Remaining limitations, if any.

3. Exploratory Data Analysis

Summarize the most important findings, including:

- Patterns.
- Trends.
- Relationships.
- Outliers.
- Interesting observations.

4. Dashboard Summary

Explain:

- The purpose of the dashboard.
- The visualizations used.
- How the dashboard answers the business questions.

5. KPI Explanation

For each KPI included in your dashboard, explain:

- What the KPI measures.
- The target value used.
- How the target was selected (historical performance, business goal, benchmark, management objective, etc.).
- Why the chosen target is appropriate.
- Whether the KPI met, exceeded, or fell below the target and what that indicates for the business.

6. Business Insights and Recommendations

Present the most important insights discovered during the analysis and provide recommendations supported by the data.

Submission Checklist

Before submitting, ensure your submission contains all of the following:

- Dataset Description (PDF)
- Cleaned Dataset (.xlsx)
- Power BI Dashboard (.pbix)
- Project Report (PDF)

Incomplete submissions may receive a grade deduction.